

ACTU 301
Life Contingencies 1

Multiple Life Functions

Lecture 5

Multiple Life Models

So far we have developed functions which depend upon the death or survival of a single life aged x . We now consider functions that depend upon the death or survival of two or more lives.

The actuarial mathematics theory that we have developed for a single life can be extended to multiple lives.

Central to the development of the functions of a single life is the random variable measuring the future lifetime of a life aged x , T_x , or its curtate counterpart K_x .

. The models that we shall discuss involve two lives.

Let's consider two lives aged x and y with future remaining lifetimes $T(x)$ and $T(y)$ respectively. We shall assume that the two future lifetime random variables are independent.

Multiple Life Statuses

The survival of the two lives is referred to as the status of interest or simply the status. There are two common types of status:

(1) The joint-life status

This is a status that survives so long as all members are alive, and therefore fails upon the first death.

That is, joint-life status is one that requires the survival of both lives. Accordingly, the status terminates upon the first death of one of the two lives.

(2) The last-survivor status is one that ends upon the death of both lives. That is, the status survives as long as at least one of the component members remains alive.

The Joint-Life Status Model

The Joint-Life Status Model

- ❖ A joint-life status of two lives (x) and (y) will be denoted by (xy) as of time 0.
- ❖ In cases where numbers are used instead of symbols x and we put colon between these numbers. For example, at time n ; we will use the notation $(x+n : y+n)$ instead of $(x+n y+n)$
- ❖ The remaining lifetime random variable of the status will be denoted by T_{xy}
- ❖ Since the status terminates upon the first death, we have
- ❖ Time until the first of the two lives fails $= T_{xy} = \min\{T_x, T_y\}$
NB: The joint status (xy) fails when (x) or (y) dies.

Joint-Life Probability Functions

The distribution function of $T=T_{xy}$ is

$$F_T(t) = \Pr(T_{xy} \leq t)$$

$$= \Pr[\min(T_x, T_y) \leq t]$$

$$= 1 - \Pr[\min(T_x, T_y) > t]$$

$$= 1 - \Pr(T_x > t, T_y > t)$$

$$= 1 - {}_t p_{xy}$$

if T_x and T_y are independent then

$$1 - \Pr(T_x > t, T_y > t) = 1 - \Pr(T_x > t) \Pr(T_y > t) = 1 - {}_t p_x \times {}_t p_y$$

Joint Survival Function of T_{xy}

For the joint survival function of T_{xy} , we have

$$S_T(t) = {}_t p_{xy} = 1 - F_T(t) = {}_t p_x \times {}_t p_y$$

Alternatively

$$S_T(t) = {}_t p_{xy}$$

$$= \Pr(T_{xy} > t)$$

$$= \Pr(\min\{T_x, T_y\} > t)$$

$$= \Pr([T_x > t] \text{ and } [T_y > t])$$

$$= \Pr(T_x > t) \Pr(T_y > t) \text{ if } T_x \text{ and } T_y \text{ are independent}$$

$$= {}_t p_x \times {}_t p_y$$

Joint-Life Probability Density Function

From this, the density function is

$$\begin{aligned} f_T(t) &= F'_T(t) = -\frac{\partial}{\partial t}(\mathcal{S}_T(t)) = -\frac{\partial}{\partial t}({}_t p_{xy}) \\ f_T(t) &= F'_T(t) = -\frac{\partial}{\partial t}({}_t p_x * {}_t p_y) \\ &= -\left({}_t p_x \frac{\partial}{\partial t} {}_t p_y + {}_t p_y \frac{\partial}{\partial t} {}_t p_x\right) \\ &= {}_t p_x({}_t p_y \mu_{y+t}) + {}_t p_y({}_t p_x \mu_{x+t}) \\ &= {}_t p_x * {}_t p_y (\mu_{x+t} + \mu_{y+t}) \end{aligned}$$

Joint Life Status Force of Mortality Function

Thus, the force of mortality $\mu_T(t)$ or $\mu_{xy}(t)$ or $\mu_{x+t:y+t}$ of T_{xy} is

$$\mu_{xy}(t) = \frac{f_T(t)}{1 - F_T(t)} = \frac{f_T(t)}{\mathcal{S}_T(t)} = \frac{{}_t p_x * {}_t p_y (\mu_{x+t} + \mu_{y+t})}{{}_t p_x * {}_t p_y}$$

$$\mu_{xy}(t) = \mu_{x+t} + \mu_{y+t} \quad \text{or} \quad \mu_{x+t:y+t} = \mu_{x+t} + \mu_{y+t}$$

The force of mortality of the joint life status is the sum of the individuals' force of mortality, when lives are independent.

Mean and Variance of T_{xy}

The complete expectation of the future lifetime of the joint-life status is

$$\begin{aligned} e^o_{xy} &= E(T_{xy}) = \int_0^\infty t f_T(t) dt = \int_0^\infty (1 - F_T(t)) dt \\ e^o_{xy} &= \int_0^\infty {}_t p_{xy} dt \end{aligned}$$

The second moment of $T(xy)$ is

$$\begin{aligned} E(T_{xy}^2) &= \int_0^\infty t^2 f_T(t) dt = 2 \int_0^\infty t {}_t p_{xy} dt \\ \text{Var}[T_{xy}] &= 2 \int_0^\infty t {}_t p_{xy} dt - [e^o_{xy}]^2 \end{aligned}$$

The curtate expectation of future lifetime of the joint-life status is

$$e_{xy} = \sum_{k=1}^{\infty} k p_{xy}$$

and the n-year curtate lifetime of the joint-life status is

$$e_{xy:n} = \sum_{k=1}^n k p_{xy}$$

Joint Life Insurances - payable at the moment of the first death

A joint life insurance (whole life insurance) is a life insurance which pays a benefit of 1 immediately upon the first to die of x and y. It is also known as a {first to die insurance}.

In this case, the random present value of future benefits is

$$v^{T_{xy}} = \exp(-\delta T_{xy})$$

which has expected present value

$$\begin{aligned}\bar{A}_{xy} &= \int_0^{\infty} e^{-\delta t} f_T(t) dt = \int_0^{\infty} e^{-\delta t} {}_t p_x {}_t p_y (\mu_{x+t} + \mu_{y+t}) dt \\ \bar{A}_{xy} &= \int_0^{\infty} v^t f_T(t) dt = \int_0^{\infty} v^t {}_t p_x {}_t p_y (\mu_{x+t} + \mu_{y+t}) dt\end{aligned}$$

Joint Life Insurances – payable at the end of the year of the first death

Let A_{xy} denotes the APV of the whole life insurance on the joint life. It pays one unit at the end of the year of the first death.

We calculate its value as follows

$$A_{xy} = E(v^{T_{xy}}) = \sum_{k=1}^{\infty} v^k {}_{k-1|}q_{xy}$$

$$\begin{aligned}\text{where } {}_{k-1|}q_{xy} &= {}_{k-1}p_{xy} - {}_k p_{xy} \\ &= {}_{k-1}p_x * {}_{k-1}p_y - {}_k p_x * {}_k p_y\end{aligned}$$

The Last-Survivor Status Model

The Last-Survivor Status Model

- ❖ A Last survivor status of two lives (x) and (y) will be denoted by $(\overline{xy})$ as of time 0.
- ❖ At time n ; we will use the notation $(\overline{x+n:y+n})$
For example, we write $T_{(\overline{34:41})}$ instead of $T_{(\overline{3441})}$
- ❖ The remaining lifetime random variable of the status will be denoted by $T_{\overline{xy}}$
- ❖ Time to the last death = $T_{\overline{xy}} = \max\{T_x, T_y\}$

Last-Survivor Probability Functions

The distribution function is

$$\begin{aligned}
 F_T(t) &= \Pr(T_{\overline{xy}} \leq t) \\
 &= \Pr(\max(T_x, T_y) \leq t) \\
 &= \Pr(T_x \leq t, T_y \leq t) \\
 &= \Pr(T_x \leq t), \Pr(T_y \leq t) \\
 &= (1 - {}_t p_x) * (1 - {}_t p_y)
 \end{aligned}$$

We write the Last-survivor survivor function as

$$S_T(t) = {}_t p_{\overline{xy}} = 1 - F_T(t) = {}_t p_x + {}_t p_y - {}_t p_x {}_t p_y$$

From this, the density function is

$$\begin{aligned}
 f_T(t) &= F'_T(t) = -\frac{\partial}{\partial t}(S_T(t)) = -\frac{\partial}{\partial t}({}_t p_{\overline{xy}}) \\
 f_T(t) &= F'_T(t) = -\frac{\partial}{\partial t}({}_t p_x + {}_t p_y - {}_t p_x {}_t p_y) \\
 &= {}_t p_x \mu_{x+t} + {}_t p_y \mu_{y+t} - {}_t p_x {}_t p_y (\mu_{x+t} + \mu_{y+t}) \\
 &= {}_t p_x \mu_{x+t} (1 - {}_t p_y) + {}_t p_y \mu_{y+t} (1 - {}_t p_x) \\
 &= {}_t p_x \mu_{x+t} {}_t q_y + {}_t p_y \mu_{y+t} {}_t q_x
 \end{aligned}$$

Last-Survivor Status Force of Mortality Function

Thus, the force of mortality

$$\mu_{\overline{xy}}(t) = \frac{f_T(t)}{1 - F_T(t)}$$

does not have a straightforward expression.

The complete expectation of $T_{\overline{xy}}$ is

The expected last survivor time is given as

$$\begin{aligned}
 e^o_{\overline{xy}} &= E(T_{\overline{xy}}) = \int_0^{\infty} {}_t p_{\overline{xy}} dt = \int_0^{\infty} ({}_t p_x + {}_t p_y - {}_t p_x {}_t p_y) dt \\
 &= e^o_x + e^o_y - e^o_{xy}
 \end{aligned}$$

Where e^0_x and e^0_y are the expected time until death for (x) and (y) respectively and e^0_{xy} is the expected time until first death of the component lives.

The curtate expectation of $T_{\overline{xy}}$ is $e_{\overline{xy}} = \sum_{k=1}^{\infty} {}_k p_{\overline{xy}}$

The temporary curtate expectation is $e_{\overline{xy}:n} = \sum_{k=1}^n {}_k p_{\overline{xy}}$

Last-survivor insurance - payable at the moment of the first death

The APV of *last-survivor insurance*, with pays a benefit of 1 immediately upon the second death, is

$$\bar{A}_{\overline{xy}} = \int_0^{\infty} e^{-\delta t} f_T(t) dt$$

$$\bar{A}_{\overline{xy}} = \int_0^{\infty} e^{-\delta t} ({}_t q_y {}_t p_x \mu_{x+t} + {}_t q_x {}_t p_y \mu_{y+t}) dt$$

$$\bar{A}_{\overline{xy}} = \int_0^{\infty} v^t ({}_t q_y {}_t p_x \mu_{x+t} + {}_t q_x {}_t p_y \mu_{y+t}) dt$$

Last-survivor insurance— payable at the end of the year of the first death

Let A_{xy} denotes the APV of the whole life insurance on the last survivor. It pays one unit at the end of the year of the first death.

We calculate its value as follows

$$A_{\overline{xy}} = E(v^{T_{\overline{xy}}}) = \sum_{k=1}^{\infty} v^k {}_{k-1|} q_{\overline{xy}}$$

where

$$\begin{aligned} {}_{k-1|} q_{\overline{xy}} &= {}_k q_{\overline{xy}} - {}_{k-1} q_{\overline{xy}} \\ &= {}_k q_x * {}_k q_y - {}_{k-1} q_x * {}_{k-1} q_y \end{aligned}$$

Comparing benefits - insurances

Type of life insurance	Single life x	Joint life status xy	Last survivor status $\overline{xy}$
Whole life - discrete	A_x	A_{xy}	$A_{\overline{xy}}$
Whole life - continuous	$\bar{A}_x$	$\bar{A}_{xy}$	$\bar{A}_{\overline{xy}}$
Term - discrete	$A_{x:\overline{n} }^1$	$A_{\overline{xy}:\overline{n} }^1$	$A_{\overline{xy}:\overline{n} }^1$
Term - continuous	$\bar{A}_{x:\overline{n} }^1$	$\bar{A}_{\overline{xy}:\overline{n} }^1$	$\bar{A}_{\overline{xy}:\overline{n} }^1$
Endowment - discrete	$A_{x:\overline{n} }$	$A_{xy:\overline{n} }$	$A_{\overline{xy}:\overline{n} }$
Endowment - continuous	$\bar{A}_{x:\overline{n} }$	$\bar{A}_{xy:\overline{n} }$	$\bar{A}_{\overline{xy}:\overline{n} }$
Pure endowment	$A_{x:\overline{n} }^1$ or ${}_nE_x$	$A_{xy:\overline{n} }^1$ or ${}_nE_{xy}$	$A_{\overline{xy}:\overline{n} }^1$ or ${}_nE_{\overline{xy}}$

Relationships Between T_{xy} and $T_{\overline{xy}}$

Recall that

$$T_{xy} = \min\{T_x, T_y\} \text{ and } T_{\overline{xy}} = \max\{T_x, T_y\}.$$

Thus, if T_{xy} assumes one of the two values T_x or T_y then $T_{\overline{xy}}$ assumes the other value. Hence, we can write

$$T_{xy} + T_{\overline{xy}} = T_x + T_y$$

$$T_{xy} * T_{\overline{xy}} = T_x * T_y$$

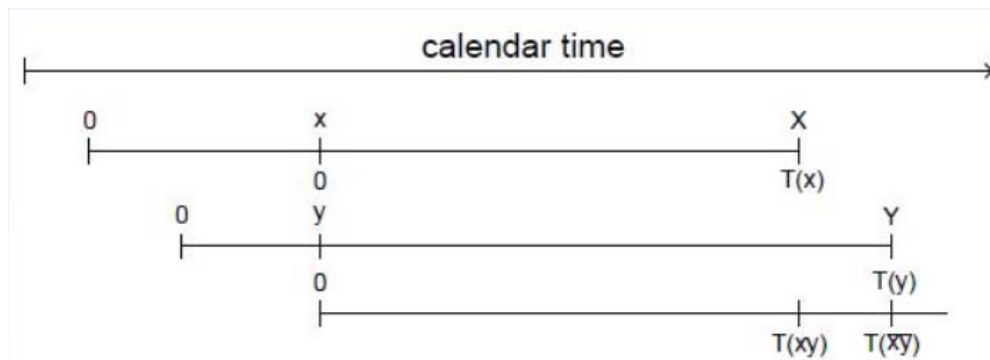
It is also worth noting, that

$$A_{xy} + A_{\overline{xy}} = A_x + A_y$$

$$\bar{A}_{xy} + \bar{A}_{\overline{xy}} = \bar{A}_x + \bar{A}_y$$

because two individual insurances on (x) and (y) have the same total pay-off as the insurances on joint life and on last survivor life, bought together.

Timeline for Two Lives



It is very useful to note that the realized value of T_x matches one of T_{xy} and $T_{\overline{xy}}$, and the realized value of T_y matches the other.

Interpretation of probabilities

Note that:

- ❖ ${}_t p_{xy}$ is the probability that both lives (x) and (y) will be alive after t years.
- ❖ ${}_t p_{\overline{xy}}$ is the probability that at least one of lives (x) and (y) will be alive after t years.

In contrast:

- ❖ ${}_t q_{xy}$ is the probability that at least one of lives (x) and (y) will be dead within t years.
- ❖ ${}_t q_{\overline{xy}}$ is the probability that both lives (x) and (y) will be dead within t years.

$$\begin{aligned} {}_u|{}_t q_{xy} &= P[(x) \text{ and } (y) \text{ are both alive in } u \text{ years but not in } u + t \text{ years}] \\ &= P[u < T_{xy} \leq u + t]. \end{aligned}$$

$$\begin{aligned} {}_u|{}_t q_{\overline{xy}} &= P[\text{at least one } (x) \text{ and } (y) \text{ is alive in } u \text{ years but both are} \\ &\quad \text{deceased after } u + t \text{ years}] \\ &= P[u < T_{xy} \leq u + t]. \end{aligned}$$

We have the following relationship for (x) and (y):

$${}_t p_x + {}_t p_y = {}_t p_{xy} + {}_t p_{\overline{xy}}$$

Example

Suppose that $T(x)$ is exponentially distributed with force of mortality $\mu_x(t)=\mu_1=0.02$, $T(y)$ is exponentially distributed with force of mortality $\mu_y(t)=\mu_2=0.015$, and that $T(x)$ and $T(y)$ are independent.

Provide expressions for:

a) ${}_t p_{xy}$

b) $\mu_{xy}(t)$

c) ${}_t \overline{p}_{\overline{xy}}$

Solution

For T_x is exponentially distributed with force of mortality $\mu_x(t)=\mu_1$,

a) ${}_t p_{xy} = {}_t p_x * {}_t p_y$

recall that ${}_t p_x = \exp(-\mu_1 t)$.

$$= \exp(-\mu_1 t) \exp(-\mu_2 t)$$

$$= \exp(-(0.02+0.015)t)$$

$$= \exp(-0.035t).$$

b) $\mu_{xy}(t) = \mu_x(t) + \mu_y(t) = \mu_1 + \mu_2 = 0.02 + 0.015 = 0.035$

c) ${}_t \overline{p}_{\overline{xy}} = {}_t p_x + {}_t p_y - {}_t p_x * {}_t p_y$

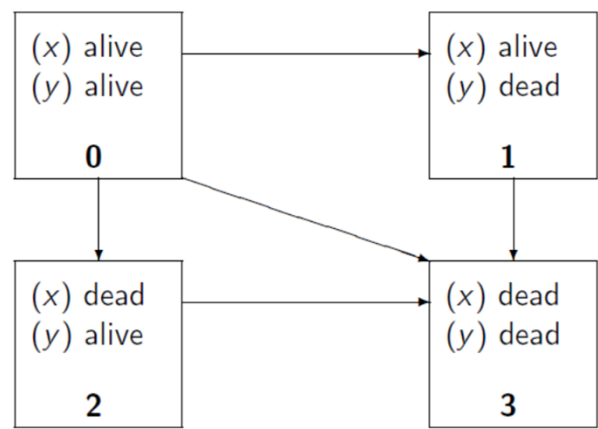
$$= \exp(-\mu_1 t) + \exp(-\mu_2 t) - \exp(-(\mu_1 + \mu_2)t)$$

$$= \exp(-0.02t) + \exp(-0.015t) - \exp(-(0.035)t)$$

“Common Shock” Model for Dependent Lives

If the model allows a transition from state 0 directly to state 3, it's known as a **common shock model**.

The common shock model allows for both lives dying at the same time, for example through a car accident.



Suppose that $T^*(x)$ and $T^*(y)$ are unobserved future lifetimes for x and y that are independent of one another. Let Z be a lifetime random variable that is common to both x and y for e.g., disasters such as earthquakes and hurricanes. Take $T^*(x)$, $T^*(y)$, Z to be mutually independent.

Let $T(x)=\min(T^*(x),Z)$ and $T(y)=\min(T^*(y),Z)$ be the observed future lifetimes for x and y . Note that they are not independent because they share the common shock random variable Z .

For convenience, assume that the distribution of Z is exponential with constant force λ .

Now, the survival function for x is

$${}_tp_x = \Pr(T(x) > t) = \Pr(\min(T^*(x),Z) > t) = \Pr(T^*(x) > t) \Pr(Z > t) = {}_tp_x^*e^{-t}$$

Similarly, ${}_tp_y = {}_tp_y^*e^{-t}$

The joint survival probability is

$$\begin{aligned} {}_tp_{xy} &= \Pr[\min(T(x),T(y)) > t] = \Pr[\min(T^*(x),T^*(y),Z) > t] \\ &= \Pr(T^*(x) > t) \Pr(T^*(y) > t) \Pr(Z > t) = {}_tp_x^*{}_tp_y^*e^{-t} \\ &= {}_tp_x{}_tp_ye^{-t} \end{aligned}$$

From this, we can also calculate the joint force of mortality

$$\begin{aligned} \mu_{xy}(t) &= -\frac{\partial}{\partial t} \ln {}_tp_{xy} \\ &= -\frac{\partial}{\partial t} (\ln {}_tp_x + \ln {}_tp_y + \ln e^{-t}) \\ &= \mu_x(t) + \mu_y(t) - \lambda \end{aligned}$$

END